Chapter 1

Database System Concept

Unit 1.5: Data Modelling: Record based logical model – Relational, Network, Hierarchical

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Learning Outcomes

The Learners will be able to:

1. Identify the relevant database model in the given situation.

Data Modelling Concepts

- A Database / data model defines the logical design of data.
- It describes the data, relationships between different parts of the data, data semantics and integrity constraints.
- A data model is underlying structure of the database.
- Relational Model is the most widely used data model.
- Three record based logical models are:
  - Hierarchical Model
  - Network Model
  - Relational Model

Hierarchical Model

- Hierarchical model was developed in 1960 and organises data into a tree-like structure.
- Its basic logical structure is represented by an upside down tree.
- In this model data storage is in the form of parent-child relationship.
- In this model each entity has only one parent but can have several children.
- At the top of hierarchy there is only one entity which is called **Root**.
- Entities which follow the root are called the **node** and the last node in the tree is called the **leaf node**.
- This model basically follows the one-to-many relationship. The following figure represents a hierarchical database.

Hierarchical Model contd..

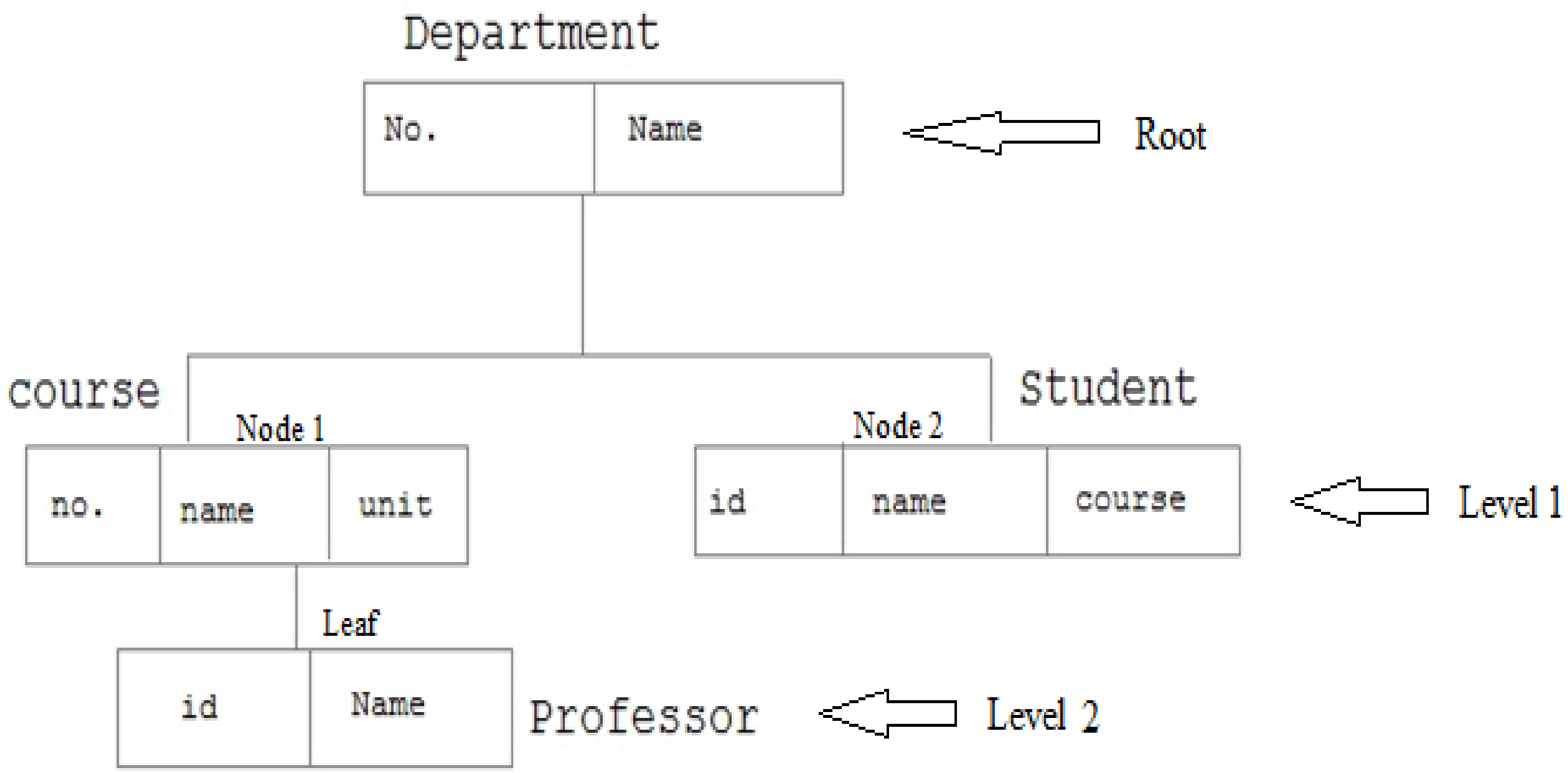


Figure: Hierarchical Model Database

Hierarchical Model contd..

Advantages:

- Simplicity in logical design
- Security of data
- Data Integrity
- Efficiency

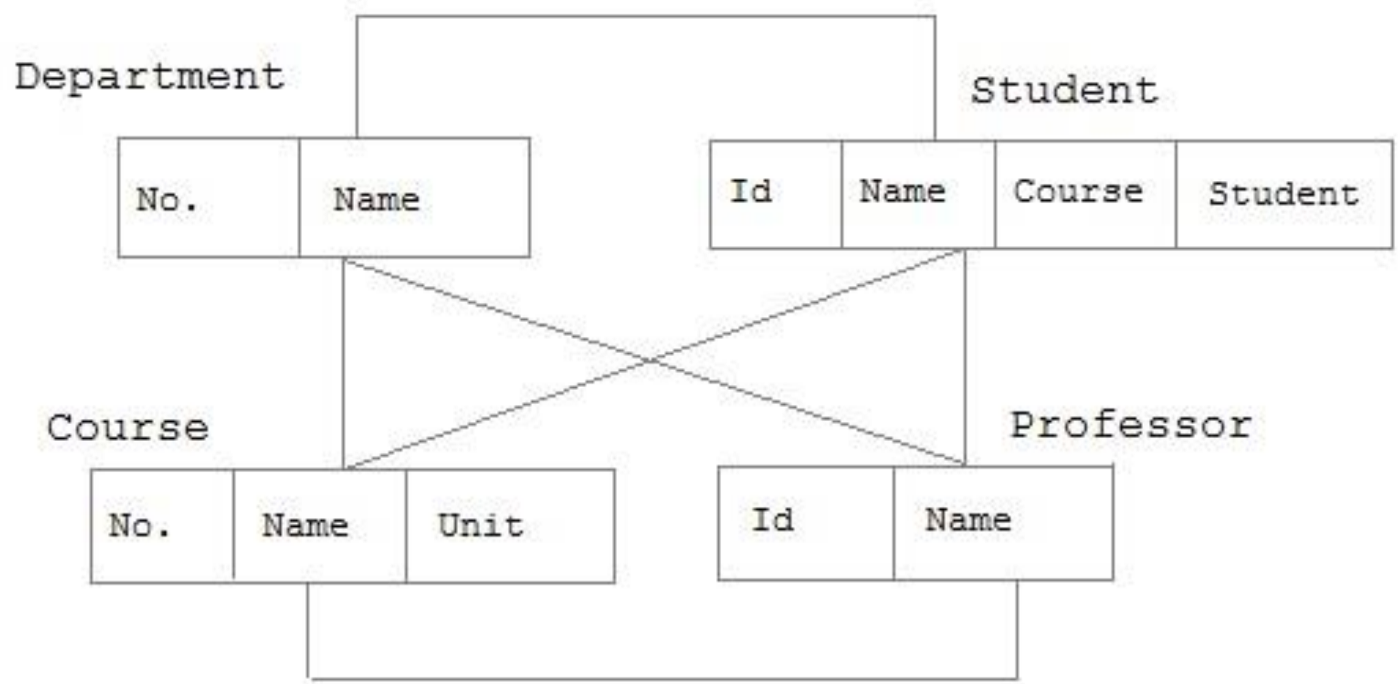
Disadvantages:

- Structural dependence
- Implementation complexity
- Data Access Complexity
- Implementation restrictions

Network Model

- Network model was developed at General Electric Company in 1964 and was standardised by CODASYL IN 1971.
- In the network model, entities are organised in a graph, in which some entities can be accessed through several path.
- Represents complex data relationships more effectively.
- Data in network model are represented by collection of records of fixed format
- Relationships among data are represented by links which can be viewed as pointers.

Network Model contd..



- In the above figure, boxes area used to represent records and lines are used to represent links between the records.
- It supports many-to-many relationship.
- The network model allows each record to have multiple parent and child records

Network Model contd..

Advantages:

- Logically simple.
- User friendly data access.
- Data integrity.
- Can handle many forms of mapping cardinality.
- Standardized approach.

Disadvantages:

- Complexity of the system.
- Structural dependence.

Relational Model

Relational model was introduced by E.F Codd in 1970.

In this model, data is organised in two-dimensional tables called **relations**. The tables or relations are related to each other.

In relational model, data is represented in terms of **tuples (rows)** & **attributes (columns)**.

Each column has a unique name.

The properties of relation(tables) are as follows:

- Values are atomic.
- Each row is unique.
- Column values are of same type.
- The sequences of column are not important.
- The sequences of rows are also not important.

Relational Model contd..

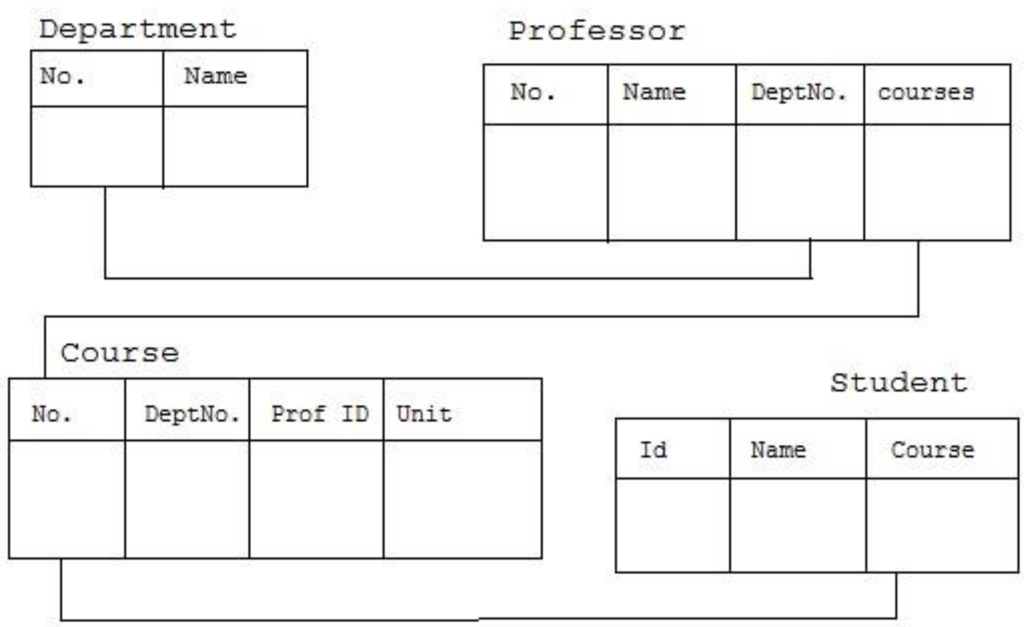


Figure: Relational Model Database

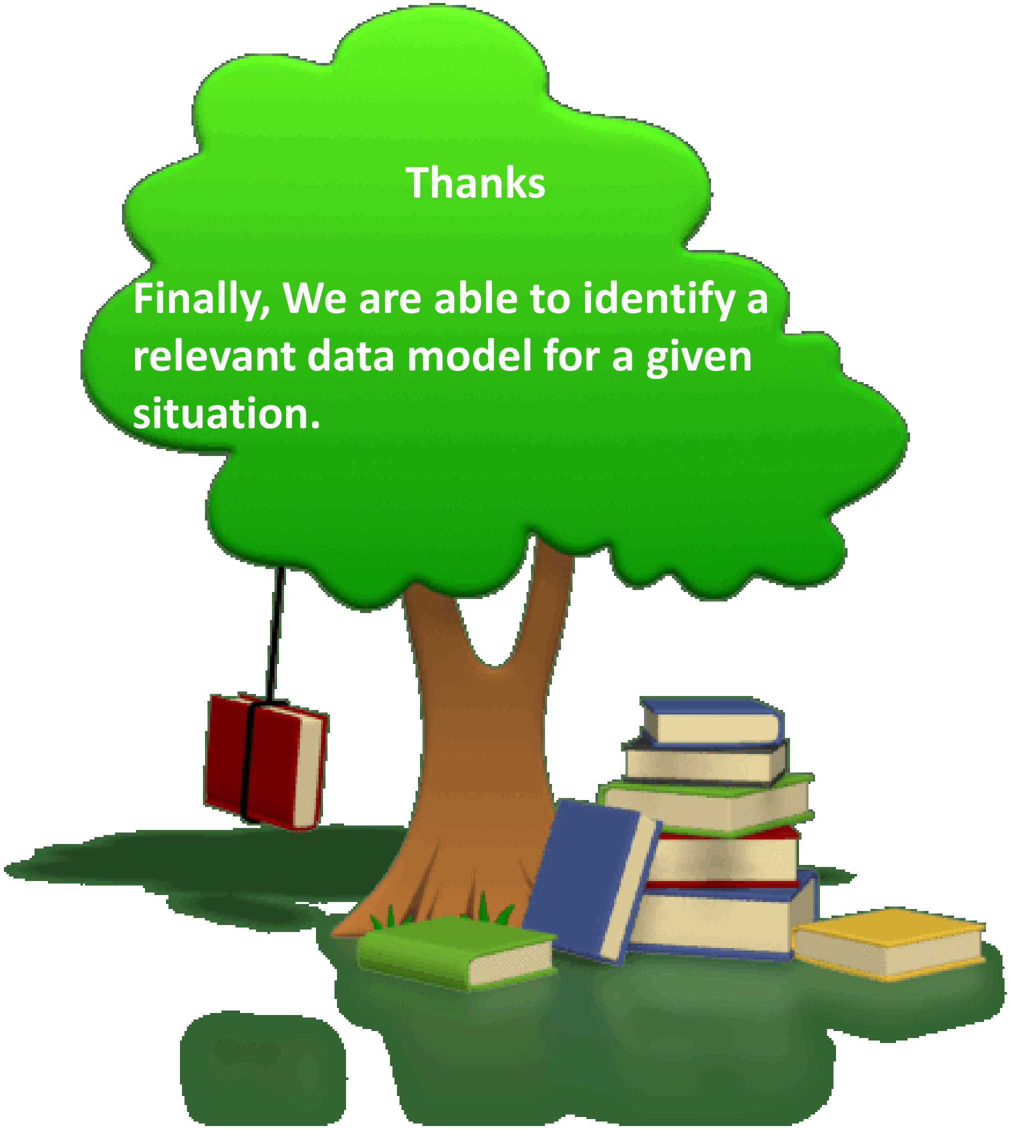
Relational Model contd..

Advantages:

- Data independence
- Simplicity
- Ease of data retrieval
- Less overhead management
- Structural independence
- Group data manipulation

Disadvantages

- Large hardware overheads
- Ease of design can lead to bad design.


Thanks
Finally, We are able to identify a relevant data model for a given situation.